

Sustained hippocampal chromatin regulation in a mouse model of depression and antidepressant action

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To better understand the molecular mechanisms of depression and antidepressant action, we administered chronic social defeat stress followed by chronic imipramine (a tricyclic antidepressant) to mice and studied adaptations at the levels of gene expression and chromatin remodeling of five brain-derived neurotrophic factor (*Bdnf*) splice variant mRNAs (I–V) and their unique promoters in the hippocampus. Defeat stress induced lasting downregulation of *Bdnf* transcripts III and IV and robustly increased repressive histone methylation at their corresponding promoters. Chronic imipramine reversed this downregulation and increased histone acetylation at these promoters. This hyperacetylation by chronic imipramine was associated with a selective downregulation of histone deacetylase (*Hdac*) 5. Furthermore, viral-mediated HDAC5 overexpression in the hippocampus blocked imipramine's ability to reverse depression-like behavior. These experiments underscore an important role for histone remodeling in the pathophysiology and treatment of depression and highlight the therapeutic potential for histone methylation and deacetylation inhibitors in depression.

Chronic stress can precipitate depression in susceptible individuals but the underlying molecular pathogenesis in the brain remains elusive. Similarly, antidepressants provide important relief in many individuals but their mechanisms of action are not completely understood^{1,2}. The hippocampus is one brain region that has been implicated in the pathophysiology of depression: human patients suffering from depression have reduced hippocampal volumes³ and rodents show stress-induced morphological changes (decreased neurogenesis and neuronal atrophy), which are reversed by antidepressant treatments^{4–6}. Brain-derived neurotrophic factor (BDNF) may have an important role in the adaptations of the hippocampus to stress and antidepressants^{1,4}. BDNF infusions in the hippocampus produce antidepressant-like effects in animal models of depression⁷, whereas mice lacking BDNF show impaired antidepressant responses⁸. Additionally, acute or chronic stress decreases BDNF expression^{9–12}, whereas chronic, but not acute, treatment with several antidepressants prevents this downregulation^{10,13}. The regulatory mechanisms mediating these changes in BDNF expression are unknown, due in part to the complicated structure of the *Bdnf* gene. The gene contains five short 5' noncoding exons in mice, each of which can be alternatively spliced to a common coding exon VI to form several mRNA transcripts, all coding for an identical BDNF protein^{14,15} (Supplementary Fig. 1 online). It is unknown whether these various transcripts differ functionally in other ways—for example, in stability or translation efficiency. However, each noncoding exon contains a unique promoter region with distinct chromatin architecture, which could modulate the expression of one splice variant over another.

Post-translational modification of histones, a form of chromatin remodeling, can alter the chromatin architecture at promoter regions by controlling their permissiveness for transcription¹⁶. In general, histone acetylation (H3 and H4 acetylation) loosens DNA-histone interactions and allows the transcriptional machinery to bind and increase transcription^{16,17}. Histone methylation, on the contrary, can correlate with either transcriptional activation (H3-lysine [K]-4, H3-K36) or repression (H3-K9, H3-K27, H4-K20), depending on the histone and amino acid residue being methylated¹⁸. Histone methylation may also facilitate DNA methylation, which causes further repression of the affected genes^{18,19}. Chromatin remodeling is increasingly recognized as a crucial mechanism in several important phenomena in the brain, including neuronal differentiation^{20,21}, neurodegeneration^{22–24}, circadian rhythm²⁵, seizure^{26,27}, memory formation^{28–32} and drug addiction³³.

Here, we studied the involvement of chromatin remodeling in the long-term neuroadaptations in depression and antidepressant action. We used an animal model of depression—namely, a chronic social defeat stress protocol, which mimics many symptoms of depression in humans^{34–39}. We show here that aspects of chronic defeat stress are reversed by chronic (but not acute) antidepressant treatment and that chronic stress and chronic antidepressant treatment are associated with reciprocal, long-lasting changes in expression levels of certain *Bdnf* splice variant mRNAs that occur in concert with lasting changes in chromatin architecture at the corresponding *Bdnf* gene promoters. In addition, we show that downregulation of *Hdac5* by chronic antidepressant treatment is critical for its therapeutic efficacy in this animal

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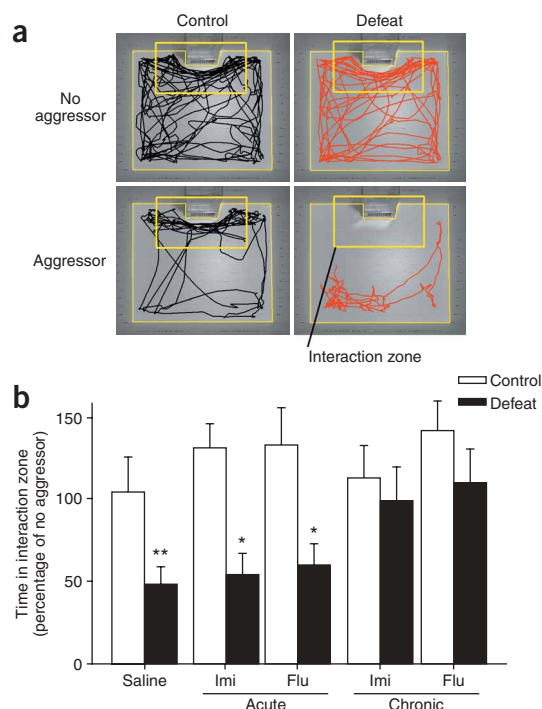


Figure 1 Social interaction and avoidance after chronic social defeat and antidepressant treatments. **(a)** Schematic representation of the interaction and avoidance test protocol. The interaction zone was defined around a cage that was either empty or contained an aggressor. The black and red lines are representative tracks of nonstressed (control) and stressed (defeat) mice in the box in the absence or presence of an aggressor. **(b)** The time spent in the interaction zone, expressed as a percent of the time spent when no aggressor was present, was measured for control and defeated mice treated with acute (1 d after 27 d of saline) or chronic (28 d) imipramine (Imi) or fluoxetine (Flu), or with saline only (28 d). In the absence of an aggressor, all four groups of mice spent similar amounts of time in the interaction zone. When an aggressor was introduced into the cage, chronically defeated mice treated with saline or acute imipramine or fluoxetine showed a ~50% reduction in their interaction time (* $P < 0.05$, ** $P < 0.01$, $n = 12$), whereas control mice (saline- and drug-treated) showed no change. In contrast, chronic imipramine or fluoxetine treatment of defeated mice reversed the defeat stress-induced decrease in interaction to a level that was significantly different from that in saline-treated defeat mice ($P < 0.01$, $n = 12$).

model. Our results establish a new role for chromatin remodeling in the long-term adaptive changes in the brain associated with depression and suggest new therapeutic mechanisms for antidepressant treatments.

RESULTS

Social avoidance induced by chronic defeat stress

Social defeat stress mimics several pathological dimensions of depression^{34,35,37–39}. To study the effect of chronic antidepressant treatment on defeated behavior, mice were subjected to chronic defeat stress for 10 consecutive days, followed by a 4-week treatment with the tricyclic antidepressant imipramine. Mice in the acute imipramine group received 27 d of saline followed by a single dose of imipramine on

the last day of injections. The chronic defeat stress protocol entailed a brief (10 min) daily exposure to a highly aggressive resident mouse, during which time the test mouse developed defeated and subordinate behavior. The aggressor and the defeated mice were then housed in the same cage, but after the 10 min of physical contact, the test and aggressor mice were physically separated by a plastic divider with holes, which allowed visual, olfactory and auditory contact for the remainder of the 24-h period. Test mice were exposed to a different resident aggressor each day for a total of 10 d. Control, nondefeated mice were housed and treated similarly but were not exposed to aggressors. At the end of the treatment, all mice were tested for their social interaction and avoidance behavior by measuring their time spent in an 'interaction zone' around a caged aggressor (Fig. 1a).

Compared to control mice, chronically defeated mice spent about 50% less time in the interaction zone when an aggressor was introduced into the cage. The administration of chronic, but not acute, imipramine reversed this avoidance behavior, increasing the interaction time so that it was close to that of the nondefeated mice (Fig. 1b). An equivalent behavioral response was seen with chronic, but not acute, fluoxetine, another type of chemical antidepressant (Fig. 1b)³⁹. Imipramine or fluoxetine injections in control mice did not have a significant effect on

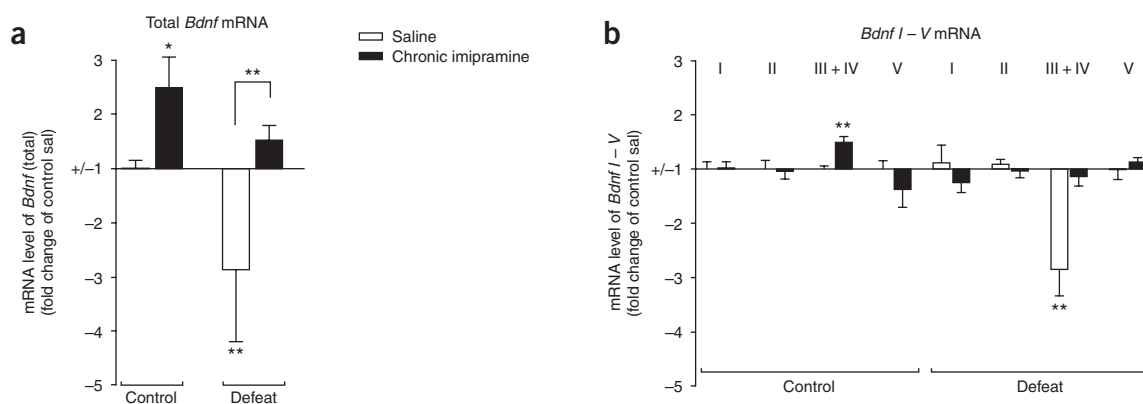


Figure 2 Differential regulation of *Bdnf* III and IV mRNA after chronic defeat stress and imipramine treatments. The mRNA levels of total *Bdnf* and *Bdnf* transcripts I–V were measured in the hippocampus of control or defeated mice receiving either saline or chronic imipramine by quantitative RT-PCR ($n = 5–6$). **(a)** Total *Bdnf* was significantly decreased ($P < 0.01$) at 4 weeks after defeat stress, compared to that in control saline-treated mice. Chronic imipramine treatment in control mice induced a significant increase in *Bdnf* compared to that in control saline-treated mice ($P < 0.05$). Chronic imipramine treatment in defeated mice reversed the stress-induced downregulation to slightly above that in the saline-treated control mice ($P < 0.01$ compared to saline-treated defeat mice). **(b)** Measurement of *Bdnf* I–V mRNAs revealed significant changes in only *Bdnf* III and *Bdnf* IV of defeated mice compared to control saline-treated mice ($P < 0.01$). In nonstressed control mice, *Bdnf* III and IV (but no other splice variants) showed a small but significant increase after chronic imipramine treatment ($P < 0.01$).

interaction time. These results confirm recent findings³⁹ and show that chronic social defeat stress may be a useful animal model of depression: namely, a long-lasting behavioral abnormality (social avoidance) can be treated with chronic (but not acute) antidepressants.

Regulation of *Bdnf* expression by chronic defeat stress

Next, we evaluated whether chronic social defeat stress and imipramine administration induce long-lasting changes in *Bdnf* gene expression in the hippocampus, based on the hypothesis that BDNF expression might be regulated through chromatin-specific events that promote the expression of distinct transcript variants. In rats, antidepressant treatments including electroconvulsive seizures (ECS) cause such promoter-specific transcriptional changes in the brain^{27,40}. Indeed, detailed mRNA analysis of whole mouse hippocampus revealed changes in only two of the five *Bdnf* transcripts. Chronic defeat stress induced an approximately threefold downregulation in total *Bdnf* mRNA levels (Fig. 2a), which was mediated by means of the decreased expression of *Bdnf* III and IV but not of any of the other variants (I, II and V) (Fig. 2b). Chronic imipramine increased total *Bdnf* expression in nonstressed mice (Fig. 2a) and this increase was again seen only for *Bdnf* III and IV mRNAs (Fig. 2b). Finally, chronic imipramine reversed the lasting *Bdnf* downregulation after chronic stress to baseline levels (Fig. 2a).

These results indicated that chronic defeat stress, even a month after its cessation, induces lasting changes in *Bdnf* expression, which are mediated specifically via *Bdnf* III and IV, and that this effect of stress is reversed by chronic treatment with imipramine. The fact that imipramine completely reversed the threefold effect of defeat stress on *Bdnf* III and IV but caused only a 50% increase in nonstressed mice suggests that imipramine's action in stressed mice was more than a simple, additive effect. Although changes in *Bdnf* III and IV transcription accounted for the entire decrease in total *Bdnf* expression in defeated mice, the magnitude of changes in these two splice variants only partially accounted for the increase in total *Bdnf* mRNA in control mice treated with imipramine. The explanation for this discrepancy is unknown (Supplementary Note online).

Regulation of histone methylation by chronic defeat stress

The notable sustained regulation of *Bdnf* III and IV expression prompted us to investigate the mechanism by which the corresponding promoters, P3 and P4, might be differentially regulated by defeat stress and antidepressant treatment to bring about the selective changes in *Bdnf* expression. We assayed the levels of several post-translational histone modifications at the different *Bdnf* promoter regions in the hippocampus using a series of chromatin immunoprecipitation (ChIP) assays of whole mouse hippocampus. We had previously standardized the ChIP technique for use in hippocampal tissue by optimizing sonication and crosslinking conditions, antibody concentration for immunoprecipitation, antibody specificity, and assay specificity and validity^{27,33} (Methods). All assays included nonimmune immunoglobulin (IgG) and no-antibody immunoprecipitations to control for the specificity of each antibody used.

We found several long-lasting changes in histone modifications with chronic social defeat stress and chronic imipramine treatment that correlated precisely with the changes we had observed for *Bdnf* III and IV mRNA regulation. We detected a strong, more than fourfold increase in H3-K27 dimethylation, a repressive histone modification marker, at the *Bdnf* P3 and P4 promoters after chronic defeat stress (Fig. 3a), with no changes apparent at the other *Bdnf* promoters (Supplementary Fig. 2 online). H3-K9 dimethylation, another histone modification that correlates with transcriptional repression, was not increased after chronic defeat stress at these promoters (Fig. 3b). Therefore, the observed downregulation of *Bdnf* III and IV correlated with a strong increase in the dimethylation of H3-K27 but not H3-K9. Chronic imipramine treatment was unable to reverse the sustained repressive methylation marker at P3 and P4 (Fig. 3a).

Several recent reports have suggested that histone methylation facilitates the methylation of DNA at specific promoter regions¹⁹. To investigate whether the strong enrichment of H3-K27 dimethylation at *Bdnf* P3 correlates with an increase in DNA methylation at CpG sites within this promoter, we assayed for the

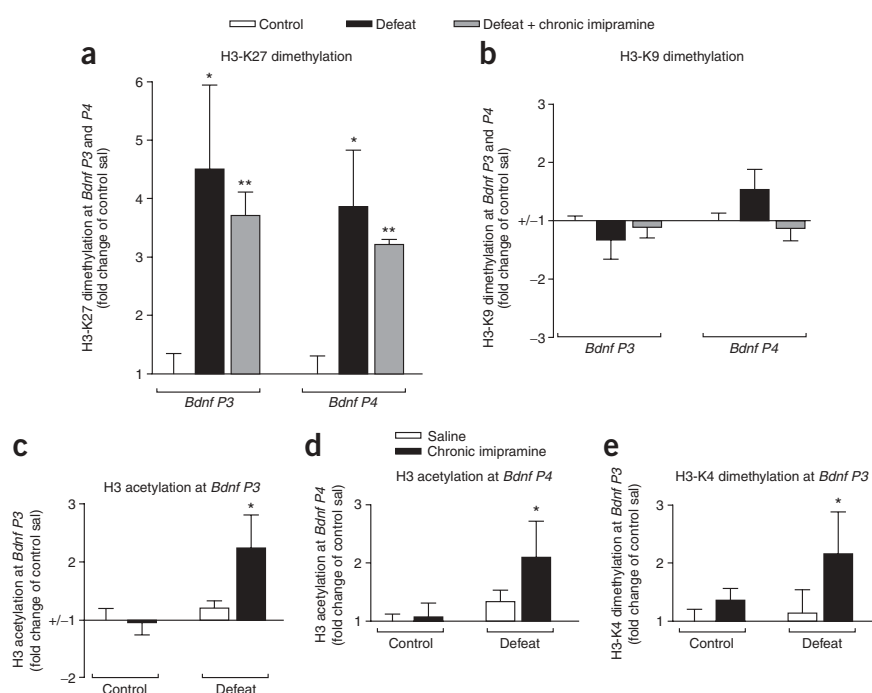
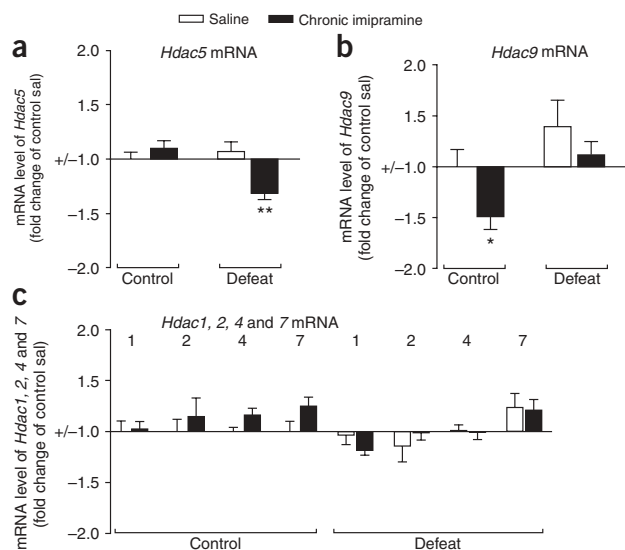


Figure 3 Stable changes in histone modifications after chronic defeat stress and imipramine treatments. ChIP assays were performed to measure the levels of several histone modifications at the five *Bdnf* promoters P1–P5 in the hippocampus after chronic defeat and chronic imipramine treatments using specific antibodies for each modification state. Levels of promoter enrichment were quantified by quantitative PCR ($n = 5–6$). (a) Histone H3-K27 dimethylation, a repressive marker of transcription, was strongly increased at *Bdnf* promoters P3 and P4 in chronically defeated mice 4 weeks after the cessation of stress ($P < 0.05$). H3-K27 dimethylation remained significantly enriched even after chronic treatment with imipramine ($P < 0.01$). (b) H3-K9 dimethylation, another repressive marker, was not altered at *Bdnf* P3 or P4. (c,d) Histone H3-K9,14 acetylation, an activating marker of transcription, was significantly enriched only after chronic imipramine treatment of defeated mice, and only at *Bdnf* promoters P3 and P4 ($P < 0.05$). (e) Histone H3-K4 dimethylation, which also correlates with transcriptional activation, was similarly enriched only in chronic imipramine-treated defeated mice at the *Bdnf* P3 promoter ($P < 0.05$). Significant changes in acetylation or methylation were not detected at other *Bdnf* promoter regions (Supplementary Figs. 2–4).



presence of methylation at all CpG sites at P3 using the sodium bisulfite treatment method. The analysis revealed that none of the CpG sites within *Bdnf* P3 nor exon III were consistently methylated after chronic defeat stress (data not shown). Therefore, at least for *Bdnf* P3, direct DNA methylation does not seem to contribute to the repression seen after chronic defeat stress.

Regulation of histone acetylation by chronic imipramine

Next, we examined if histone modifications known to correlate with transcriptional activation^{17,41} are altered after chronic social defeat stress and chronic treatment with imipramine. We discovered that, in defeated mice, chronic imipramine induces long-lasting, more than twofold hyperacetylation at histone H3 selectively at *Bdnf* P3 and P4, the promoters driving de-repression of *Bdnf* III and IV mRNAs under these conditions (Fig. 3c,d). In contrast, no changes were observed at the other *Bdnf* promoters whose transcripts are not regulated by imipramine (Supplementary Fig. 3 online). Chronic imipramine did not produce hyperacetylation at *Bdnf* P3 and P4 in control (nondefeated) mice, which indicates that imipramine produces this modification only in mice previously exposed to defeat stress. Dimethylation of H3-K4, another histone modification that correlates with transcriptional activation¹⁶, showed similar patterns of

Figure 4 Expressional analysis of class I and II *Hdacs*. The mRNA levels of *Hdacs* 1, 2, 4, 5, 7 and 9 in the hippocampus were measured by quantitative PCR in nonstressed and defeated mice receiving either saline or imipramine ($n = 6$). (a) *Hdac5* mRNA was significantly downregulated in chronically defeated mice receiving chronic imipramine ($P < 0.01$). This experiment was repeated twice, yielding similar results. *Hdac5* mRNA was not changed in any of the other experimental groups. (b) *Hdac9* mRNA was downregulated only in nonstressed mice treated with chronic imipramine ($P < 0.05$). (c) No significant changes were observed for *Hdacs* 1, 2, 4 and 7 with any of the described treatments.

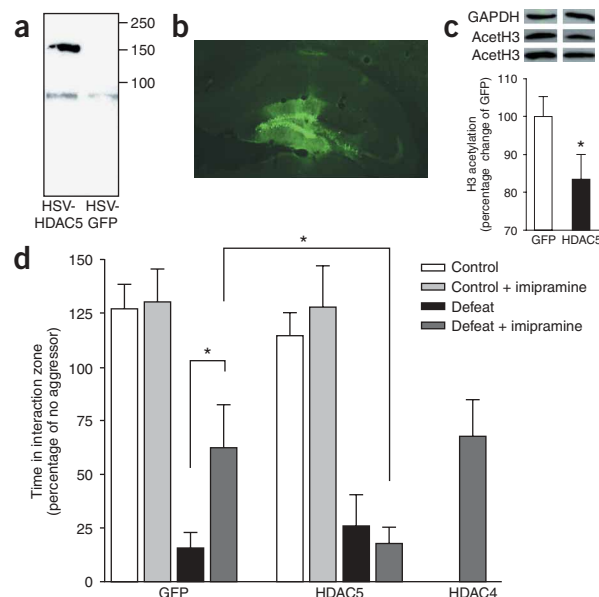
enrichment at *Bdnf* P3, again only after chronic imipramine treatment of defeated mice (Fig. 3e, Supplementary Fig. 3). Acetylation of H4, which likewise correlates with transcriptional activation, did not show detectable changes at *Bdnf* promoters under any of these treatment conditions (Supplementary Fig. 4 online).

To investigate the mechanisms underlying these lasting and selective changes in histone modifications, particularly H3 hyperacetylation, we hypothesized that levels of specific HDACs may be regulated by chronic antidepressant treatment. We measured expression levels of both class I and class II *Hdacs* (1,2,4,5,7 and 9) in the hippocampus of control and defeated mice with and without subsequent treatment with imipramine. We observed significant changes of expression in only two isoforms: *Hdac5* and *Hdac9* (Fig. 4). *Hdac9* mRNA levels were decreased in nonstressed mice receiving chronic imipramine (Fig. 4b), whereas *Hdac5* mRNA levels were decreased in chronically stressed mice treated with chronic imipramine (Fig. 4a). Acute imipramine treatment did not produce these effects (data not shown) and, similarly, defeat stress alone had no influence on levels of *Hdac* expression (Fig. 4).

Regulation of defeat behavior by histone acetylation

The observation that chronic imipramine downregulates *Hdac5* expression in socially defeated mice but not in control mice led us to speculate that this effect may be important for the therapeutic efficacy of the drug. To study this possibility directly, we overexpressed HDAC5 in the dentate gyrus region of the hippocampus by use of herpes simplex virus (HSV) viral vectors (Fig. 5). This region was specifically targeted because it shows the greatest degree of *Bdnf* mRNA regulation¹². Human *HDAC5* cDNA was subcloned

Figure 5 Viral-mediated overexpression of HDAC5 blocks the ability of chronic imipramine to reverse behavioral deficits of chronic defeat stress. (a) HSV-HDAC5 (which also expresses GFP) was transfected into PC12 cells, and HDAC5 protein levels were measured by Western blotting. The immunoblot shows strong viral-mediated overexpression of HDAC5 (160 kDa) protein *in vitro*. (b) The same vector was injected into the dentate gyrus of the hippocampus; GFP expression was revealed by immunohistochemistry. (c) Western blots of two representative mice showing that viral-mediated overexpression of HDAC5 (compared to HSV-GFP infection) decreased total levels of acetylated H3 (K9,14), compared to GAPDH levels, in the dentate gyrus ($P < 0.05$; $n = 6$). (d) HSV-HDAC5 was injected into the dentate gyrus of defeated and control mice on day 25 of a 28-d chronic treatment with imipramine or saline. HDAC4 was overexpressed only in defeated mice treated with chronic imipramine. Social interaction and avoidance was measured 3 d after infection. Compared to HSV-GFP, HSV-HDAC5 did not significantly alter the interaction time of control mice given saline, control mice treated with imipramine or defeated mice given saline ($n = 8-9$). However, HSV-HDAC5 blocked imipramine's efficacy in reversing social avoidance after defeat stress ($n = 10$; $P < 0.05$). No effect was seen with HSV-HDAC4 ($n = 7$).



into a bicistronic HSV–green fluorescent protein (GFP) virus⁴² and then packaged for infection. First, we confirmed that the virus overexpresses *HDAC5* *in vitro* and *in vivo* and that this overexpression decreases overall levels of H3 acetylation (Fig. 5a–c). We then proceeded to infect the dentate gyrus bilaterally with HSV-HDAC5-GFP or HSV-GFP (as a control). We also overexpressed HDAC4, another class II HDAC that is expressed in the hippocampus but shows no regulation by imipramine (Fig. 4c). The virus was introduced on day 25 after imipramine or saline treatment of defeated or control mice, and the social interaction and avoidance behaviors of the mice were measured on the third day after infection, when the expression levels of this HSV virus are maximal⁴³. The avoidance behavior of the mice was then scored by measuring the amount of time spent in the interaction zone around the caged aggressor mouse, as described earlier (Fig. 5d).

Neither HSV-HDAC5-GFP nor HSV-GFP affected the behavior of nonstressed mice, which showed a high degree of social interaction, as seen in uninjected controls. This was true for mice treated chronically with imipramine or saline. Similarly, mice that were stressed but did not receive imipramine showed the expected increase in avoidance behavior, regardless of whether they received HSV-HDAC5-GFP, HSV-GFP or no viral injections. Finally, mice injected with HSV-GFP that were defeated but received chronic imipramine resembled uninjected controls in that they showed more social interaction and less avoidance. In notable contrast, however, HDAC5 overexpression had a profound effect on defeated mice treated with imipramine, where it completely blocked the ability of imipramine to reverse avoidance and increase social interaction (Fig. 5d). These mice showed very robust avoidance behavior, similar to that of defeated mice not treated with imipramine. These behavioral effects of HDAC5 overexpression occurred in the absence of any change in total locomotor activity (Supplementary Table 1 online). In contrast, the overexpression of HDAC4 did not affect the ability of imipramine to reverse defeat behavior (Fig. 5d). These results directly implicate the downregulation of HDAC5 in the hippocampus in the antidepressant-like activity of imipramine. Many genes presumably mediate this behavioral effect of HDAC5 and *Bdnf* is one putative candidate: compared to the expression of GFP, the viral-mediated overexpression of HDAC5 in the hippocampus significantly blunted imipramine's ability to induce the expression of *Bdnf* exons III and IV ($23 \pm 3\%$ reduction with HDAC5 versus GFP overexpression, $n = 5$, $P < 0.05$).

In contrast, we found that sodium butyrate, one of whose functions is to act as a nonspecific HDAC inhibitor, given systemically twice daily for 21 d after social defeat showed a strong trend for an antidepressant-like effect (interaction time with aggressor as a percent of controls: saline, $21 \pm 11\%$; butyrate, $50 \pm 16\%$; $n = 8$, $P < 0.06$). The lack of a more robust effect may be due to the fact that sodium butyrate, at maximal tolerated doses, causes only modest increases in histone acetylation in brain³³ and that it acts broadly throughout the brain on all HDACs and also on other targets. A more definitive test of the antidepressant potential of HDAC inhibition will require the development of more selective HDAC inhibitors: in particular, a selective inhibitor of HDAC5 that is specifically implicated in mediating imipramine's effects in our social defeat model.

DISCUSSION

The forced-swim and tail-suspension tests are among the most commonly used methods to examine depression-like behavior and antidepressant activity in laboratory mice. Although these tests can predict antidepressant activity at least for existing mechanisms of action⁴⁴, they have been criticized because acute antidepressants are active in the tests

even though chronic exposure is required for their clinical utility in humans. The chronic defeat stress model overcomes this limitation in that the social avoidance induced in the test can be treated with chronic, but not acute, antidepressant administration. Another advantage of the social defeat protocol is that it involves a social form of chronic stress, which may be relevant to stress-induced psychopathology in humans. The reversal of defeat-induced social avoidance in mice by antidepressants suggests that this behavioral pathology may be relevant to human depression. Nevertheless, this model may be relevant to other psychiatric phenomena as well, such as fear, anxiety, social phobia and post-traumatic stress disorder. Further work is needed to more completely characterize the clinical relevance of the social defeat protocol.

Using this social defeat model, we found several lasting modifications in the hippocampus at the level of chromatin remodeling at the *Bdnf* gene, which precisely paralleled the changes in *Bdnf* gene expression. We discovered that histone dimethylation at H3-K27 is strongly enriched after chronic social defeat stress and that this modification is extremely long-lasting, being present at the *Bdnf* P3 and P4 promoters even a month after the cessation of stress. This signifies that chronic stress can mark a repressive state that cannot be easily reversed⁴⁵. Indeed, chronic treatment with imipramine did not significantly alter the hypermethylation level of H3-K27. This finding could have profound implications for the treatment of depression. Current antidepressants are effective in ameliorating symptoms of depression but not in curing the disorder, and in many patients (at least half) symptoms reappear after the discontinuation of treatment^{1,2}. This underscores the importance of better understanding the pathological effect of chronic stress on the brain and the need for better therapeutic agents to reverse these effects. Our findings suggest that histone hypermethylation may represent a stable stress-induced molecular scar in the hippocampus and perhaps elsewhere and that the search for newer antidepressant agents will identify those that are more effective in demethylating histones at repressed genes.

Although histone methylation was not reversed by antidepressant treatment, chronic imipramine did induce long-lasting H3 hyperacetylation at the *Bdnf* P3 and P4 promoters in defeated mice. Further investigation of the mechanism for this stable modification revealed that chronic imipramine downregulates a specific HDAC (HDAC5) in these defeated mice. In addition, the overexpression of HDAC5 completely blocked the antidepressant efficacy of imipramine in the social defeat protocol. These findings suggest that the downregulation of HDAC5 has an essential role in the therapeutic action of imipramine and that specific HDAC5 inhibitors (which unfortunately are not yet available) might possess a new antidepressant efficacy.

Although H3 acetylation was upregulated by chronic imipramine in defeated mice at *Bdnf* P3, acetylation of H4, another modification that marks transcriptional activation, remained unchanged at this promoter. However, we observed reductions in H4 acetylation after an acute stress challenge, when H3 acetylation was unaffected (data not shown). This interesting phenomenon of an apparent switch from the acute regulation of H4 to the chronic regulation of H3 has been observed previously in two other systems: chromatin changes induced in the hippocampus by acute ECS involve the acetylation of H4, whereas those induced by chronic ECS involve the acetylation of H3 (ref. 27); similarly, chromatin changes induced in the striatum by acute and chronic cocaine involve the acetylation of H4 and H3, respectively³³. Further research is needed to understand the mechanisms of this interesting switch.

A notable finding of the current study is that imipramine downregulates HDAC5 and hyperacetylates H3, but only in mice that were previously subjected to chronic social defeat stress, with no effect seen in normal controls. This is an intriguing observation because

imipramine and other antidepressants are generally thought to have no major effect in normal humans and to exert a mood-elevating effect only in depressed individuals (**Supplementary Note**). **The mechanism underlying this selective effect of imipramine has long been unknown, and our findings could provide a unique experimental system in which to understand this action.** For example, it is possible that the hyperacetylating effects of imipramine are greatly exaggerated in defeated mice because the *Bdnf* gene is highly repressed as a result of histone methylation. In other words, **imipramine may downregulate the expression of HDAC5 and hyperacetylate affected genes only in an off-balance system, such as after chronic stress.** A related question concerns the mechanism by which chronic imipramine regulates the expression of HDAC5. We would assume that chronic imipramine exerts this effect by means of its **perturbation of the brain's noradrenergic and serotonergic systems, although** further work is needed to study the molecular basis of this regulation.

Together, our data suggest a model by which chronic stress induces repression and chronic imipramine induces de-repression of the *Bdnf* gene in the hippocampus (see model in **Supplementary Fig. 5**). Under normal conditions, the *Bdnf* gene is expressed at some basal level. Chronic defeat stress induces the specific dimethylation of histone H3 at residue K27 at the *Bdnf* gene, which persists long after the end of stress. This chromatin modification leads to a more 'closed' chromatin state and thereby mediates the stable repression of the *Bdnf* gene. The regulation of histone acetylation, DNA methylation and HDAC5 expression are not affected after chronic stress alone, which corroborates the idea that the repression of the *Bdnf* gene is mediated mainly via histone methylation. Chronic imipramine induces hyperacetylation of H3 at the *Bdnf* promoter after chronic defeat stress, an effect that seems to be mediated at least in part by means of the downregulation of HDAC5 expression. Hyperacetylation of the promoter partially overcomes its methylation-induced repression and leads to a more 'open' chromatin state at the *Bdnf* promoter. This causes de-repression of the *Bdnf* gene and contributes to imipramine's antidepressant activity. Because histone methylation is a very thermodynamically stable modification and is thus not easily reversed⁴⁵, an opening of the chromatin by means of histone hyperacetylation provides a more plausible mechanism of de-repression than that via histone demethylation.

We view the *Bdnf* gene as a prototypical example of how chronic stress and chronic imipramine presumably produce many additional long-lasting changes in gene expression in the hippocampus and other brain regions at the level of chromatin remodeling. Future genome-wide experiments will be aimed at identifying these other genes and providing new leads toward understanding the pathophysiology and treatment of depression. Although many questions remain about the mechanisms by which chronic stress and chronic imipramine cause changes in chromatin architecture at the *Bdnf* and other genes, the results of the present study provide fundamentally new information concerning the detailed molecular mechanisms underlying the deleterious effects of stress on the brain and their reversal by chronic antidepressant treatment. More generally, our results provide further support for the notion that chromatin remodeling is an important mechanism controlling long-term adaptive changes in the brain associated with complex psychiatric conditions.

METHODS

Animal housing and injections. Adult male BL6/C57 mice (9–11 weeks old; Jackson Laboratory) were used in all experiments. Mice were single-housed and maintained on a 12-h light-dark cycle with access to food and water *ad libitum*. All animal procedures were carried out in accordance with Institutional Animal Care and Use (IACUC) guidelines. Mice were injected daily intraperitoneally

(i.p.) with imipramine (20 mg per kg body weight), fluoxetine (20 mg per kg body weight), sodium butyrate (200 mg per kg body weight) or saline. Mice were analyzed 24 h after their last injection.

Chronic social defeat stress. Defeat stress was carried out using a method similar to that reported recently³⁹. BL6/C57 test mice were exposed to a different CD1 aggressor mouse each day for 10 min over a total of 10 d. During the brief exposure, all test mice showed signs of stress and subordination, including vocalization, flight response and a submissive posture. After the 10 min of contact, test mice were separated from the aggressor: the test mice were placed in an adjacent compartment of the same cage, separated by a plastic divider with holes, where they were exposed to chronic stress in the form of threat for the next 24 h. Control test mice were housed in equivalent cages but with members of the same strain, which changed daily. Twenty-four hours after the last session, all mice were housed individually for 4 weeks, during which time they received daily (i.p.) injections of imipramine, fluoxetine, sodium butyrate or saline. Mice in acute treatment groups received saline for 27 d and one dose of the drug on the last day of injections. The day after the last injection, we tested the long-term behavioral consequences of the chronic defeat stress using a measure of interaction and avoidance toward one of the aggressors used during the defeat procedure. Mice were placed in a new arena with a small animal cage at one end, and their movement was tracked for 2.5 min in the absence of the aggressor, followed by 2.5 min in the presence of the caged aggressor. We obtained the duration in the interaction zone (**Fig. 1a**) and other measures using Ethovision 3.0 software. See **Supplementary Methods** online for further details.

mRNA analysis. Whole hippocampus was extracted for RNA quantification from the following groups of mice 24 h after their last injection and 2 d after their last behavioral test: chronic defeat stress plus 4 weeks of saline, chronic defeat stress plus 4 weeks of chronic imipramine, nonstressed controls plus saline and nonstressed controls plus chronic imipramine. RNA was processed as described previously²⁷. Primers were designed complementary to each *Bdnf* noncoding exon I–V to assay for the level of expression of each individual transcript, or to exon VI to measure levels of total *Bdnf* mRNA (**Supplementary Table 2** online).

ChIP assays. We used a recently published ChIP technique with minor modifications^{27,33}. Primers were designed around the putative promoter regions of *Bdnf* P1–5, upstream of each exon I–V (**Supplementary Table 2**). We used antibodies to acetylated H3 (K9,14), acetylated H4 (K5,8,12,16) and dimethylated H3 (K4), H3 (K9) and H3 (K27) (Upstate Biotechnology). The real-time polymerase chain reaction (PCR) ChIP data was analyzed identically to the mRNA data, using the $\Delta\Delta C_t$ method, except that ChIP data were normalized to 'input' rather than to glyceraldehyde-3-phosphate dehydrogenase (GAPDH). As with all mRNA analyses, we used whole hippocampus to perform all ChIP experiments. As a control, we measured histone modifications at the promoters of the *Tubb* and *Syp* genes, which are expressed in adult hippocampus but whose expression levels are reportedly unchanged in this region after ECS (refs. 27,46,47). We also performed several additional controls to confirm the specificity of our ChIP assays (**Supplementary Methods**).

DNA methylation. DNA methylation analysis was performed according to published methods and Chemicon's DNA methylation Kit protocols, with a few modifications^{48–50}. To extract DNA, hippocampal tissue was incubated at 55 °C overnight in lysis buffer (10% SDS, 1 M Tris pH 7.5, 0.5 M EDTA, 5 M NaCl, 1.5 M dithiothreitol (DTT), 100 mM spermidine) containing proteinase K (10 $\mu\text{g ml}^{-1}$). The tissue was then incubated at 37 °C for 2 h with RNase A (10 $\mu\text{g ml}^{-1}$). DNA was extracted using a mixture of phenol, chloroform and isoamyl alcohol (25:24:1), precipitated with ethanol and resuspended in 50 μl of a solution containing 10 mM Tris (pH 7.4) and 0.1 mM EDTA. The DNA was then treated with sodium bisulfite to convert all nonmethylated cytosines into thymidines. For sodium bisulfite treatment and desalting, the Chemicon kit and protocol were used without any changes. Modified DNA was then amplified using primers specific for bisulfite-treated DNA, which did not include any CpG sites where possible methylation could be present (**Supplementary Table 2**). The following PCR amplification conditions were used: 95 °C for 2 min, 1 cycle; 95 °C for 1 min, 50 °C or 54 °C for 2 min, 72 °C for

3 min, 5 cycles; 95 °C for 45 s, 50 °C or 54 °C for 1.5 min, 72 °C for 1.5 min, 25 cycles; 72 °C for 6 min, 1 cycle. The amplified DNA products were cloned directly using TOPO-TA cloning kit (Invitrogen) and transformed into TOP10 competent cells. Five different colonies from each DNA amplification reaction were then analyzed for possible methylated CpG sites using direct sequencing from the TOPO plasmids containing the insert.

Viral-mediated gene transfer. *HDAC5* and *HDAC4* cDNAs (gift of E. Olson, University of Texas Southwestern) were subcloned into a published bicistronic HSV-GFP virus vector⁴², which was then packaged as described⁴³. Viral-mediated *HDAC4* and *HDAC5* overexpression levels were confirmed *in vitro* by Western blotting (using antibodies from Upstate) and *in vivo* by real-time PCR (RT-PCR) using specific primers (Supplementary Table 2) for human *HDAC4* or *HDAC5* (Fig. 5a; other data not shown). Viral injections *in vivo* were carried out according to standard methods (Supplementary Methods).

Statistical analysis. Two- or three-way ANOVA and Bonferroni post-test analyses were used to determine statistical significance for all social interaction and avoidance behavioral data (**P* < 0.05; ***P* < 0.01). For all chromatin immunoprecipitation and mRNA data, fold changes relative to control saline were determined using the $\Delta\Delta Ct$ method²⁷; a mean fold change ($2^{-\Delta\Delta CtAVE}$) value along with an s.e.m. ($(abs(((2^{-\Delta\Delta CtAVE} \times 2^{-\Delta\Delta CtSEM}) - (2^{-\Delta\Delta CtAVE} / 2^{-\Delta\Delta CtSEM}))) / 2))$ value were determined; the $\Delta\Delta Ct$ values from each data set were used in two-tailed paired *t*-tests (which were adjusted for multiple comparisons) to determine statistical significance (**P* < 0.05; ***P* < 0.01). All values included in the figure legends represent mean $\pm$ s.e.m.

Note: Supplementary information is available on the Nature Neuroscience website.

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COMPETING INTERESTS STATEMENT

The authors declare that they have no competing financial interests.

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